

Last time:

Want to solve equations of form

$$\frac{dX_t}{dt} = \mu(X_t) + \sigma(X_t) \times \text{"noise"}$$

- "noise" = Brownian motion

$$\bullet W_t, W_t - W_s \sim N(0, t-s),$$

• independent increments

• continuous paths. ←

(Not differentiable)

- Quadratic Variation:

$$V_t^{\pi}(W) = \sum (W_{t_i, n, t} - W_{t_{i-1}, n, t})^2$$

Then $V_t^{\pi}(W) \rightarrow t$
in $L^2(\mathbb{P}) \& \mathbb{P}$.

§3: Stochastic Integration "Ito's"

$$X_t - X_0 = \int_0^t \mu(X_s) ds + \int_0^t \sigma(X_s) dW_s$$

How to interpret?

More generally: how to understand

$$\int_0^t \varphi_s dW_s,$$

(φ_t) is some stochastic process.

Simple Example:

$$\varphi_t = W_t. \text{ Compute } \int_0^1 W_s dW_s$$

$$\approx \sum_{k=0}^{N-1} W_{t_k} \Delta W_k,$$

N fixed, $\Delta t = 1/N$, $t_k = k \Delta t$,

$$\Delta W_k = W_{t_{k+1}} - W_{t_k}$$

$$= \sum_{k=0}^{N-1} W_{t_k} (W_{t_{k+1}} - W_{t_k})$$

$$a(b-a) = \frac{1}{2} b^2 - \frac{1}{2} a^2 - \frac{1}{2} (b-a)^2$$

$$= \sum \left(\frac{1}{2} W_{t_{k+1}}^2 - \frac{1}{2} W_{t_k}^2 \right)$$

$$- \frac{1}{2} \sum (W_{t_{k+1}} - W_{t_k})^2$$

$$= \frac{1}{2} W_{t_N}^2 - \frac{1}{2} W_{t_0}^2 - \frac{1}{2} \sum_{k=0}^{N-1} (W_{t_{k+1}} - W_{t_k})^2 \quad (7)$$

$$= \frac{1}{2} W_1^2 - \frac{1}{2} W_0^2 - \frac{1}{2} \sum_{k=0}^{N-1} (W_{t_{k+1}} - W_{t_k})^2$$

→ quadratic variation

"Classically" expect:
 $(\frac{1}{2}W_t^2)' = W_t dW_t$

But Thm 2.6 $\Rightarrow$

$$\langle W \rangle_1 = \lim_{N \rightarrow \infty} \sum_{i=1}^N (W_{t_{i+1}} - W_{t_i})^2 = 1$$

$$\text{so } \sum W_{t_k} (W_{t_{k+1}} - W_{t_k}) \rightarrow \frac{1}{2} (W_1^2 - W_0^2)$$

We'd like to call the limit the $-\frac{1}{2}$ stochastic integral:

$$\int_0^t W_s dW_s = \frac{1}{2} (W_t^2 - W_0^2) - \frac{1}{2} t$$

However

Why did we choose to approx.

W_t over $[t_i, t_{i+1})$ by W_{t_i} ?

Could take $W_{t_{i+1}}$ instead?

$$\text{then } b(b-a) = \frac{1}{2} b^2 - \frac{1}{2} a^2 + \frac{1}{2} (b-a)^2$$

$$\begin{aligned} \Rightarrow \sum_{k=0}^{N-1} W_{t_{k+1}} (W_{t_{k+1}} - W_{t_k}) &= \frac{1}{2} W_1^2 - \frac{1}{2} W_0^2 + \frac{1}{2} \sum_{k=0}^{N-1} (W_{t_{k+1}} - W_{t_k})^2 \\ &\rightarrow \frac{1}{2} (W_1^2 - W_0^2) + \frac{1}{2} \end{aligned}$$

Not $-\frac{1}{2}$

• In the second case, integrand "peeks ahead".

For now: Assume we take left end point

$\rightarrow$ Note: assume ϕ_t is adapted
 i.e. depends only on (W_s) up to time t

§3.1 Construction for "simple" processes.

Def 3.1: The set $\mathcal{U}$ of progressively measurable processes is the set of functions $\varphi: \mathbb{R}_+ \times \Omega \rightarrow \mathbb{R}$

$$(w, t) \mapsto \varphi_t(w) = \varphi_t$$

(i) φ_t is adapted

$$(ii) \mathbb{E} \left[\int_0^\infty \varphi_t^2 dt \right] < \infty$$

[+ measurability conditions]

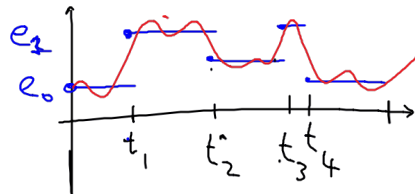
If a prog. meas. process is of the form:

$$\varphi_t(w) = \sum_{j=0}^{n-1} e_j(w) \mathbb{1}_{[t_j, t_{j+1})}(t)$$

for fixed times $0 \leq t_0 \leq t_1 \leq \dots \leq t_n$

then φ_t is simple, $\varphi \in \mathcal{S}$

Note: $e_j(w)$ known at t_j .



Defn 3.2:

Let $\varphi \in S$. Then the stochastic integral of

φ_t w.r.t. W_t is:

$$\int_0^\infty \varphi_s dW_s = \sum_{j=0}^{n-1} \varphi_{t_j} (W_{t_{j+1}} - W_{t_j})$$

More generally

$$\int_u^v \varphi_s dW_s = \int_0^\infty \varphi_s \underbrace{\mathbb{1}_{[u,v)}}(s) dW_s$$

simple if φ is.

§3.2 Properties of integral of simple processes.

Lemma 3.3: If $\varphi, \psi \in S, \alpha, \beta \in \mathbb{R}$

- $0 \leq s \leq u \leq t \in \mathbb{R}_+ \cup \{\infty\}$
- (i) $\int_s^t \varphi_u dW_u = \int_s^u \varphi_u dW_u + \int_u^t \varphi_u dW_u$
 - (ii) $\int_s^t (\alpha \varphi_r + \beta \psi_r) dW_r = \alpha \int_s^t \varphi_r dW_r + \beta \int_s^t \psi_r dW_r$
 - (iii) $\mathbb{E} \left[\int_s^t \varphi_u dW_u \right] = 0 = \mathbb{E} \left[\int_s^t \varphi_u dW_u \middle| \mathcal{F}_s \right]$
 - (iv) $t \mapsto \int_0^t \varphi_u dW_u$ is continuous and adapted.

(v) (Itô Isometry)

$$\mathbb{E} \left[\left(\int_0^t \varphi_u dW_u \right)^2 \right] = \mathbb{E} \left[\int_0^t \varphi_u^2 du \right]$$

Proof of (v): φ simple $\Rightarrow \varphi_u = \sum \varphi_{t_i} 1_{[t_i, t_{i+1})}$

$$\begin{aligned} \mathbb{E} \left[\left(\int_0^t \varphi_u dW_u \right)^2 \right] &= \mathbb{E} \left[\left(\sum \varphi_{t_i} (W_{t_{i+1}} - W_{t_i}) \right)^2 \right] \\ &= \sum_{i,j} \mathbb{E} \left[\varphi_{t_i} \varphi_{t_j} \Delta W_i \Delta W_j \right] \end{aligned}$$

• consider $i \neq j, i < j$. tower prop.

$$\begin{aligned} \mathbb{E} [\varphi_{t_i} \varphi_{t_j} \Delta W_i \Delta W_j] &= \mathbb{E} [\mathbb{E} [\varphi_{t_i} \varphi_{t_j} \Delta W_i \Delta W_j | \mathcal{F}_{t_j}]] \\ &= \mathbb{E} [\varphi_{t_i} \varphi_{t_j} \Delta W_i \mathbb{E} [\Delta W_j | \mathcal{F}_{t_j}]] \\ &= 0 \end{aligned}$$

indep $\varphi_j = \mathbb{E}[\Delta W_j]$

• $i = j$

$$\begin{aligned} \mathbb{E} [\varphi_{t_j}^2 \Delta W_j^2] &= \mathbb{E} [\varphi_{t_j}^2 \mathbb{E} [\Delta W_j^2 | \mathcal{F}_{t_j}]] \\ &= (t_{j+1} - t_j) \mathbb{E} [\varphi_{t_j}^2] \\ &= \mathbb{E} \left[\int_{t_j}^{t_{j+1}} \varphi_{t_j}^2 du \right] \end{aligned}$$

So $\mathbb{E} \left[\left(\int_0^t \varphi_u dW_u \right)^2 \right]$

$$\begin{aligned} &= \mathbb{E} \left[\sum_{i=1}^{n-1} \varphi_{t_i}^2 (t_{i+1} - t_i) \right] \\ &= \mathbb{E} \left[\int_0^t \varphi_u^2 du \right] \quad \square \end{aligned}$$

§3.3 Itô isometry & Stochastic Integration

Key idea: Above: map from
 S to $L^2(\mathbb{P})$
 $= \{X: \Omega \rightarrow \mathbb{R} \mid \mathbb{E}[X^2] < \infty\}$

$L^2(\mathbb{P})$ is a complete metric space $\Rightarrow$ use this to define limits.

Thm 3.4 Let $\varphi \in V$. Then
 $\exists \varphi^n \in S$ s.t.

$$\lim_{n \rightarrow \infty} \mathbb{E} \left[\int_0^\infty (\varphi_n - \varphi^n)^2 du \right] = 0 \quad (10)$$

Then $I := \lim_{n \rightarrow \infty} \int_0^\infty \varphi^n dW_u$
exists in $L^2(\mathbb{P})$, and is indep. of
the choice of φ^n .

Pf: (sketch...) (10) in notes.

Now suppose we are given such a
sequence.

Then:

$$\mathbb{E} \left[\int_0^\infty (\varphi_n^m - \varphi_n^n)^2 du \right] \leq 2 \mathbb{E} \left[\int_0^\infty (\varphi_n^m - \varphi_n^m)^2 du \right] + 2 \mathbb{E} \left[\int_0^\infty (\varphi_n^n - \varphi_n^n)^2 du \right]$$

So a Cauchy sequence. $\rightarrow 0$ as $n, m \rightarrow \infty$

By the Itô isometry, $I^n = \int_0^\infty \varphi_n^n dW_u$

$$\mathbb{E}[(I^n - I^m)^2] \rightarrow 0 \text{ as } n, m \rightarrow \infty$$

Hence $I^n \rightarrow I \in L^2(\mathbb{P})$,

since $L^2(\mathbb{P})$ complete.

Can see unique by interleaving sequences. $\square$

Defⁿ 3.5:

For $\varphi \in \mathcal{U}$, the (Itô) integral
is defined as the limit of
the I^n 's given above.
(Any φ^n 's satisfying (10).)

Cor 3.6 (Itô Isometry)

If $\varphi \in \mathcal{U}$

$$\mathbb{E}\left[\int_0^\infty \varphi_u^2 du\right] = \mathbb{E}\left[\left(\int_0^\infty \varphi_u dW_u\right)^2\right]$$

In fact all of Lemma 3.3 hold.

Stratonovich:

$$\int_0^t \varphi_u \circ dW_u$$

midpoint, not Left point

$$\int_0^t \varphi_u \circ dW_u = \int_0^t \varphi_u dW_u + \text{process}$$

∴

can identify this